

A Hybrid Framework for Demand Forecasting

Abstract

This paper presents a hybrid framework for demand forecasting, bridging the gap between traditional statistical modeling and context-aware predictive analytics. By integrating domain-specific contextual features, we demonstrate a significant improvement in predictive accuracy.

1. Introduction

Demand forecasting is a foundational element of supply chain efficiency. While historical sales data provides a baseline, it often lacks the nuance required to navigate volatile market conditions. This paper introduces a hybrid approach that combines the Random Forest Regressor with external environment-aware weighting.

2. Methodology & Results

Our research utilized a dataset of 76,000 records. We achieved a baseline MAPE of 43.51 percent. By implementing our hybrid framework, which incorporates weather and epidemic status as predictive weights, we reduced the MAPE to 39.37 percent.

Appendix: Hybrid Feature Engineering

```
weather_map = {'Sunny': 1.2, 'Cloudy': 1.0, 'Rainy': 0.9, 'Snowy': 0.8}
epidemic_map = {0: 1.0, 1: 0.6}
```